



| AA-5200 - 22 On ground | Spring-2 | Visualization, Feedback and Dissemination |

Instructor: Juan Ignacio Gonzalez, Ph.D.	Course Meeting Time: Tuesdays & Thursdays, 9:30 to 11:30 AM.
Email: juanignacio.gonzalez@slu.edu	Meeting Location: Morrissey Hall, Room 0600.
Office Location: Brouster Hall, office 254	Pre-requisites / Co-requisites: None.
Office Hours: Mondays & Wednesdays, 10:00 AM to 12:00 PM.	Course Assistant: Ganesh Kota, Master of Science, Analytics. ganesh.kota@slu.edu

COURSE CATALOG DESCRIPTION

This course will expose students to visualization and presentation techniques designed for the interpretation of data, improved comprehension, communication, and decision-making. Students will use current software tools to analyze data, design interfaces, and create interactive visualizations, shareable reports, and presentation applications. Topics will include data and image models, design and evaluation of reporting structures, maps and mapping, document collections, object interaction, feedback processes, and scientific and business simulations.

COURSE LEARNING OUTCOMES

By the end of this course, students will be able to:

1. Apply a methodology to guide any descriptive data analysis or data mining project.
2. Identify key issues and actions needed to integrate and prepare a dataset for analysis.
3. Use advanced visualizations to examine, scrutinize, and validate analyses driven by research questions.
4. Apply automated statistical methods and descriptive models and interpret the results to support the analysis.
5. Create reports and dashboards of data analysis using visualization methods.
6. Be proficient in using and leveraging a cutting-edge data visualization platform.

COURSE GOALS

This course will help students to:

- a) Integrate learning across disciplines to enhance understanding.
- b) Critically evaluate and incorporate information and its sources.
- c) Utilize industry-leading software to perform complex and meaningful evaluations and presentations.
- d) Demonstrate writing as a mode of inquiry and use storytelling to present the results of descriptive analyses.



REQUIRED MATERIALS

- **Textbook:** Arnold, J. (2022). *Learning Microsoft Power BI*. Transforming Data into Insights. Sebastopol: CA. O'Reilly Media, Inc. 1st Ed. ISBN: 978-1-098-11284-4.
- **Other materials:**
 1. Deckler, G. & Powell, B. (2022). *Mastering Microsoft Power BI: Expert techniques to create interactive insights for effective data analytics and business intelligence*. Birmingham: OK. Packt Publishing Ltd. 2nd. Ed. ISBN 978-1-80181-148-4
 2. Practice exercises: Microsoft Power BI Data Analyst. Course PL-300T00-A. Self-directed learning: <https://learn.microsoft.com/en-us/training/courses/pl-300t00#course-syllabus>

TIME EXPECTATIONS AND GRADING

Students should be prepared to spend an average of 12 hours of work per week, for this course. The course's accelerated nature demands that students stay on top of work and communicate with the instructor or course assistant as soon as any problems or concerns arise.

COURSE DELIVERY/METHODOLOGY

This course is delivered on-ground. Students enrolled in this course are responsible for participating in the following ways.

- **Class attendance.** Students are required to attend scheduled classes to enhance knowledge and skill acquisition throughout the term.
- **Student Introduction.** The "Introduce Yourself" discussion post will be used to give your professor and your classmates a little background information about yourself. *For this reason, it is mandatory for you to complete them by the dates and hours shown on the Course Calendar.*
- **Readings.** Assigned readings should be read at least once, with multiple readings recommended for difficult material. Optional reference documents are available for this course, your job, your capstone project, or work in the Analytics profession.
- **In-class presentations.** Teams will solve, submit, and present selected in-class cases each week, reinforcing models, concepts, and practices reviewed during our sessions. Teamwork will be peer-reviewed, and some teams will present in class. **Remote work is not permitted for this activity. Only students who attend the corresponding sessions and actively participate will receive points for this activity.**
- **Self-assessments.** Self-assessments are quick tests with up to six multiple-choice questions in Canvas, based on in-class exercises and prior activities (readings,



videos). **These assessments take place at the end of the first session each week, count toward the final grade, and will only be credited to students who attend class and submit them during the session. No remote completion or submission is permitted.**

- **Mid-term and final evaluations.** These summative evaluations assess the student's comprehension, application, and synthesis of knowledge acquired across the term.
- **Final Project:** A final workbook project consisting of a case study based on a dataset, selected by the student.

All activities are to be submitted individually, except for the in-class presentations.

COURSE GRADING SCALE

Component	Points	Total	%	Letter Grade Assignment
Self-assessments (7)	7x4	28	12%	94 to 100 = A
Mid-term evaluation (1)	1x35	35	16%	90 to 93 = A-
Final exam (1)	1x47	47	21%	87 to 89 = B+
In-class presentations (6)	6x5	30	13%	84 to 86 = B
(80% corresponds to the grade assigned to submission and 20% comes from the individual evaluation of team members)				80 to 83 = B-
Reflection essay	1x5	5	2%	77 to 79 = C+
Final Power BI Analysis Project		80	36%	70 to 76 = C
<i>Intermediate deliverables (2)</i>	2x15	30	14%	60 to 69 = D
<i>Final deliverable (1)</i>	1x50	50	22%	0 to 59 = F
	Total	221	100%	

FQ - The grade of FQ should be given to those students who had ceased attending/participating in a course and as a result earned a failing grade.



LATE OR MISSING ASSIGNMENTS

Please refer to the course calendar for the due dates of assignments. All assignments are due by 11:59 PM on the week and day listed in the calendar and instructions. Any submission made after this time will be considered late. Late assignments will incur a penalty of a 4.17% deduction for each hour past the due date. *Students do not need special permission or authorization to submit late deliverables; however, such submissions will be subject to the Late submission policy described ahead.*

ATTENDANCE POLICY

Success in this course depends on your attendance, and missing a full week of class, whether one or multiple sessions, during an eight-week term can seriously impact your learning. Failure to attend for two weeks or more puts you at serious risk of failing the course. Medical documentation may be necessary in cases of serious illness or medical conditions which impact your ability to participate in the course for an extended period. The instructor reserves the right to excuse absences and accept late work. In the event of a chronic medical condition or disability, students should contact the Dean of Students and/or the Office of Disability Services to explore potential accommodation.

ACADEMIC INTEGRITY

Academic integrity is the commitment to and demonstration of honest and moral behavior in an academic setting. Since the mission of the University is “the pursuit of truth for the greater glory of God and for the service of humanity,” acts of integrity are essential to its very reason for existence. Therefore, the University considers academic integrity a matter of serious importance. Academic integrity is the foundation of the academic assessment process, which in turn sustains the ability of the University to certify to the outside world the skills and attainments of its graduates. Adhering to the standards of academic integrity allows all members of the University to contribute to a just and equitable learning environment that cultivates moral character and self-respect. The full University-level Academic Integrity Policy can be found here:

<https://www.slu.edu/provost/policies/academic-and-course/academic-integrity-policy.pdf>

POLICY ON THE USE OF ARTIFICIAL INTELLIGENCE AND GENERATIVE AI TOOLS

I encourage the use of generative AI tools to support research activities, guide knowledge searches, and enhance learning. While AI is a valuable tool for enhancing efficiency and broadening perspectives, it should complement, not replace, your personal work, creativity, and intellectual development. Any evidence of direct copy-pasting will be addressed accordingly.

STUDENT SUCCESS CENTER

The Student Success Center (SSC) supports students in reaching their goals in and out of the classroom. Providing a variety of resources, the Student Success Center houses both the Center for Accessibility and Disability Resources (CADR) and Academic Support, which



includes Tutoring, Supplemental Instruction, University Writing Services, and Student Success Coaching. The Student Success Center can be accessed by the Busch Student Center, Suite 331, and students can make an appointment with any SSC resource via EAB Navigate. To learn more about the Student Success Center and its resources, please visit our website: <https://www.slu.edu/life-at-slu/student-success-center/index.php>

DISABILITY ACCOMMODATIONS

Students with a documented disability who wish to request academic accommodation must formally register their disability with the University. Once successfully registered, students also must notify their course instructor that they wish to use their approved accommodations in the course.

Please contact the Center for Accessibility and Disability Resources (CADR) to schedule an appointment to discuss accommodation requests and eligibility requirements. Most students on the St. Louis campus will contact CADR, located in the Student Success Center and available by email at accessibility_disability@slu.edu or by phone at 314-977-3484. Once approved, information about a student's eligibility for academic accommodation will be shared with course instructors by email from CADR and within the instructor's official course roster. Students who do not have a documented disability but believe they may have one are also encouraged to contact CADR. Confidentiality will be observed in all inquiries.

TITLE IX

Saint Louis University and its faculty are committed to supporting our students and seeking an environment that is free of bias, discrimination, and harassment. If you have encountered any form of discrimination based on sex, including sexual harassment, sexual assault, stalking, domestic or dating violence, we encourage you to report this to the University. Discrimination based on sex includes discrimination based on assigned sex at birth, sex characteristics, pregnancy and pregnancy related conditions, sexual orientation, and gender identity. If you speak with a faculty member about an incident that involves a Title IX matter, that faculty member must notify SLU's Title IX Coordinator that you shared an experience relating to Title IX. This is true even if you ask the faculty member not to disclose the incident. The Title IX Coordinator will then be available to assist you in understanding all your options and in connecting you with all resources on and off campus.

If you are pregnant or experiencing a pregnancy related condition, the Title IX Coordinator can assist you in understanding your rights and options as well as provide supportive measures.

Anna Kratky is the Title IX Coordinator at Saint Louis University (DuBourg Hall, room 36; anna.kratky@slu.edu; 314-977-3886). If you wish to speak with a confidential source, you may contact the counselors at the University Counseling Center at 314-977-TALK or make



an anonymous report through SLU's Integrity Hotline by calling 1-877-525-5669 or online at <http://www.lighthouse-services.com/slu>. To view SLU's policies, and for resources, visit this [site](#).

WELLNESS

With our Jesuit commitment to *cura personalis*, “caring for the whole person,” the University sees your academic success as connected to your health and well-being and provides resources to support your holistic wellness. All students experience stressors and challenges at some point, and seeking support is both normal and beneficial. Such challenges may be the result of academic concerns (such as those related to particular assignments or content in a course), or they may be more personal in nature (such as concerns related to relationships, mental health, medical issues, loss, identities, alcohol or drugs, housing or food security, finances, or local/world events, among other things). If you experience these or other difficulties that are impacting your well-being and/or academic work, please consider seeking support from the resources available to you.

UNIVERSITY COUNSELING CENTER

The University Counseling Center (UCC) offers free, short-term, solution-focused counseling to Saint Louis University undergraduate and graduate students. UCC counselors are highly trained clinicians who can assist with a variety of issues, such as adjustment to college life, troubling changes in mood, and chronic psychological conditions. To make an appointment for a wellness consultation, call 314-977-8255 (TALK), or visit the clinic on the second floor of Wuller Hall. For after-hours needs, please press #9 after dialing the clinic number.

BASIC NEEDS SECURITY

Students experiencing food insecurity, housing insecurity, and any other challenges that are impacting their personal and/or academic wellbeing are encouraged to contact the Dean of Student's Office for support. Students can submit an intake form, email deanofstudents@slu.edu, or call 314-977-9378 to connect with their office. Students may also communicate directly with their instructors about any challenges they are experiencing to receive support and resource referrals.

STUDENT COURSE RECORDING POLICY

Saint Louis University prohibits recording and transmission of classroom lectures and discussions by students unless written permission from the class instructor has been obtained and all students in the class as well as guest speakers have been informed that audio/video recording may occur. Recordings, course materials, and lecture notes may not be exchanged or distributed for commercial purposes, for compensation, or for any other purpose other than study by students enrolled in the class. Public distribution of such materials may constitute copyright infringement in violation of federal or state law, or



University policy. Violation of this policy may subject a student to disciplinary action under the University's Student Handbook.



AA5200 Spring 2025-2. Calendar and Roadmap. On ground.

W	SESSION	TOPICS	READINGS		EVALUATIONS	DISCUSSIONS	IN-CLASS CASE SOLUTION	FINAL PROJECT	OTHER ACTIVITIES
			Textbook and/or readings	DUE DATE (Readings)					
0	Pre-work	None	None	17-Mar	-	-	-	-	AW0: Download and install Power BI Desktop, and obtain the Textbook. Due: 3/17
1	Intro to Data Visualization and Power BI	1.1 Introduction to the course 1.2 The Data Analysis process and DPAD 1.3 Introduction to Data Analysis and Data Analyst Roles 1.4 Getting Started with Power BI and Its Integration 1.5 Enhancing Business Efficiency with Power BI	Ch 1	17-Mar	-	DW1: Introduce yourself in the d. forum. Due: 3/24	INC1: In-class case solution and presentation 1. Due: 3/20 by the end of class	-	SA1: Respond to the self-assessment on first class of the week, at the end of the session. Due: 3/18
2	Getting data in Power BI	2.1 Data acquisition and connection 2.2 Data management and storage 2.3 Performance issues	Ch 2	24-Mar	-	-	INC2: In-class case solution and presentation 2. Due: 3/27 by the end of class	-	SA2: Respond to the self-assessment on first class of the week, at the end of the session. Due: 3/25
3	Clean, transform, and load data in Power BI	3.1 Data cleaning and transformation 3.2 Data profiling and evaluation 3.3 Query management and transformation 3.4 Advanced data editing and customization	Ch 3	31-Mar	-	-	INC3: In-class case solution and presentation 3. Due: 4/3 by the end of class	FP1: Final Project Deliverable 1. Due: 4/7	SA3: Respond to the self-assessment on first class of the week, at the end of the session. Due: 4/1
4	Modeling data in Power BI	4.1 Designing Semantic Models 4.2 Managing relationships 4.3 Creating and structuring data tables	Reading 1	7-Apr	MTE: Mid Term Evaluation. Due: 4/10	-	-	-	SA4: Respond to the self-assessment on first class of the week, at the end of the session. Due: 4/8
5	Adding measures to Power BI models	5.1 Adding and managing measures 5.2 Creating and using calculated columns 5.3 Creating calculated tables 5.4 Understanding row and column Context 5.5 Using DAX basic calculations	Ch 5	14-Apr	-	-	INC5: In-class case solution and presentation 5. Due: 4/17 by the end of class	-	SA5: Respond to the self-assessment on first class of the week, at the end of the session. Due: 4/15
6	Creating Visualizations in Power BI	6.1 Design of reports 6.2 Report Structure and objects 6.3 Selection of appropriate Visual Types 6.5 Design and configuration of Report filters and slicers	Ch 4	21-Apr	-	-	INC6: In-class case solution and presentation 6. Due: 4/24 by the end of class	FP2: Final Project Deliverable 2. Due: 4/28	SA6: Respond to the self-assessment on first class of the week, at the end of the session. Due: 4/22
7	Designing Reports and Dashboards in PowerBI	7.1 Use of advanced Filtering Techniques 7.2 Application of consumption-time Filtering 7.3 Enhancing Report designs for user experience 7.4 Dashboards	Ch 6 & 7	28-Apr	-	-	INC7: In-class case solution and presentation 7. Due: 5/1 by the end of class	-	SA7: Respond to the self-assessment on first class of the week, at the end of the session. Due: 4/29
8	Orientation and work on Final Project	No new topic will be offered	None	5-May	FE: Final Evaluation. Due: 5/8	-	-	FPR: Final Project Report and RE: Reflection Essay. Both: Due: 5/10	-

Dates are subject to change, the points on this Roadmap/Calendar supersede the syllabus. ACTIVITIES ARE ALWAYS DUE IN CANVAS BY 11:59 PM ON THE DUE DATE, EXCEPT FOR THE SELF-ASSESSMENT, which must be completed at the end of each session, or any other due date and time the instructor may indicate in class or through a Course Announcement.